# Gradient Descent and Optimization

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

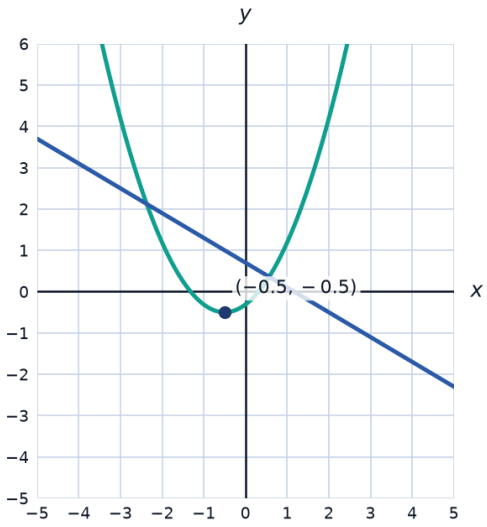
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

[Partial Derivatives & the Gradient · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

**2** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

[The Multivariable Chain Rule, Directional Derivatives & Gradient Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.




**3** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

[Multivariable Optimization: Critical Points, Hessians & Lagrange Multipliers · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

**4** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Use the method of Lagrange multipliers to find the maximum and minimum values of $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 25$.

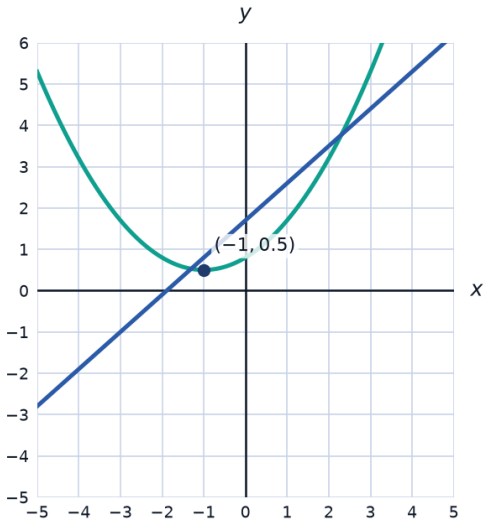
**5** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Level 3 · Advanced Find the extreme values of $f(x, y) = x^2 + 2y^2$ on the circle constraint $g(x, y) = x^2 + y^2 = 1$ using the Method of Lagrange Multipliers.

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**6** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Let $z = x^2y - y^3$, where $x(t) = \cos(2t)$ and $y(t) = \sin(2t)$. Use the multivariable chain rule to find $\frac{dz}{dt}$ when $t = 0$.




**7** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Level 2 · Medium Find the maximum rate of change of $f(x, y) = 3x^2y - y^3$ at the point $(1, 2)$, and determine the unit vector in which it occurs.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

Level 1 · Foundations Find $f_x(2, 1)$ and $f_y(2, 1)$ for the function $f(x, y) = \ln(x^2 + 3y^2)$.

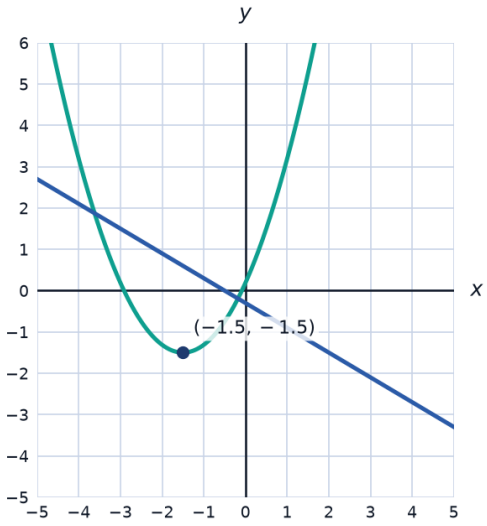
**9** [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

[Differential Equations · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

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**10** **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

[Foundations & Precalculus Review · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.




**11** **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

[The Fundamental Theorem · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

**12** **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

[Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

**13** **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

[Limits & Continuity · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

**14** **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

[Multiple Integrals · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

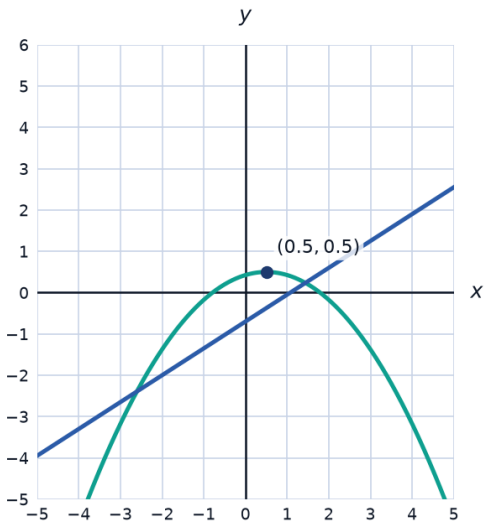
**15** **[Challenging]** • **Target: Gradient Descent & Optimization Practice Exercises** • **Cross-Lesson: 4.01 Conditional Probability and Bayes Rule**

[Parametric & Polar · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

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**16** **[Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood**

[Sequences & Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.




**17** **[Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS**

[Techniques of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

**18** **[Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification**

[Vector Calculus · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

**19** **[Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

[AP Statistics Exam Format & Study Design · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

**20** **[Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI**

[Unit 1 · Exploring One-Variable & Two-Variable Data · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input $t$ when $g(t) = 72$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In gradient descent, a weight currently has value \( w = 10 \). If the gradient of the loss is \( \frac{\partial L}{\partial w} = 4 \) and the learning rate is \( \eta = 0.1 \), find the updated weight \( w' \).

**a.** \( 10.40 \)

**b.** \( 9.60 \)

**c.** \( 6.00 \)

**d.** \( 9.90 \)

**22** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

Consider the scalar loss function \( L(w) = 2w^2 - 8w + 5 \). Find the analytical gradient \( \nabla L(w) \) evaluated at \( w = 2 \).

**a.** \( \nabla L(2) = 16 \)

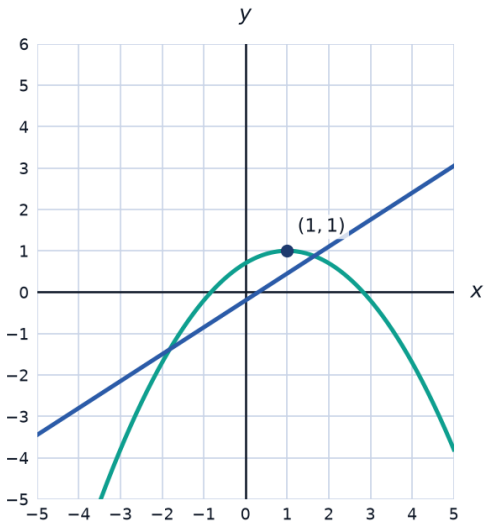
**b.** \( \nabla L(2) = 8 \)

**c.** \( \nabla L(2) = 0 \)

**d.** \( \nabla L(2) = -4 \)

**23** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

A 2D loss surface is defined as $L(w_1, w_2) = 2w_1^2 + 3w_2^2$. Find the gradient vector $ablaL$ at the point $(w_1, w_2) = (2, -1)$.




**a.** $ablaL = [10, -8]^T$

**b.** $ablaL = [8, 6]^T$

**c.** $ablaL = [4, -3]^T$

**d.** $ablaL = [8, -6]^T$

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**24** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In training a language model token predictor, a binary token occurs $k = 3$ times across $n = 10$ independent trials. What is the Maximum Likelihood Estimate $\hat{p}_{\text{MLE}}$ of the token probability?

**a.** $\hat{p}_{\text{MLE}} = 0.300$

**b.** $\hat{p}_{\text{MLE}} = 0.450$

**c.** $\hat{p}_{\text{MLE}} = 0.150$

**d.** $\hat{p}_{\text{MLE}} = 0.700$

**25** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

A machine learning diagnostic test has sensitivity $P(\text{Positive}|\text{Defect}) = 0.90$. If the defect rate in production is $P(\text{Defect}) = 0.05$, find the joint probability of a defect and a positive test result $P(\text{Defect} \cap \text{Positive})$.

**a.** 0.0450

**b.** 0.9500

**c.** 0.8500

**d.** 0.0050

**26** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

For a bivariate feature dataset, $\text{Cov}(x, y) = 12$, $\text{Var}(x) = 4$, $\bar{x} = 5$, and $\bar{y} = 25$. Compute the OLS regression slope $\hat{\beta}_1$ and intercept $\hat{\beta}_0$.

**a.** $\hat{\beta}_1 = 0.3333333333333333,\ \hat{\beta}_0 = 10$

**b.** $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 10$

**c.** $\hat{\beta}_1 = 3,\ \hat{\beta}_0 = 25$

**d.** $\hat{\beta}_1 = 4,\ \hat{\beta}_0 = 5$

**27** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

A spam classification model identifies $TP = 30$ genuine spam emails and $FP = 10$ legitimate emails falsely flagged as spam. What is the precision of this model?

**a.** 75.0%

**b.** 60.0%

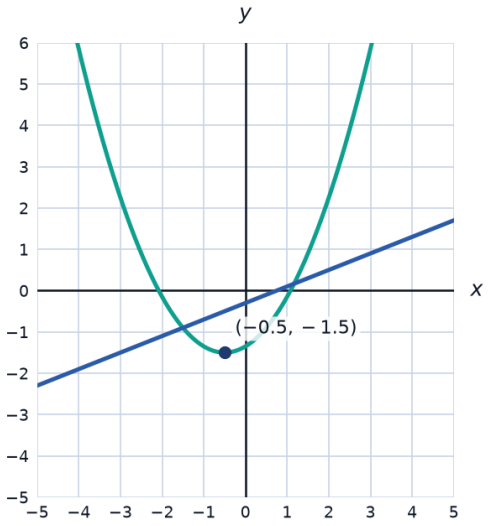
**c.** 25.0%

**d.** 85.7%

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**28** [Foundational] • Target: Gradient Descent & Optimization Practice Exercises

In evaluating a binary classifier, the confusion matrix shows $TP = 45$, $TN = 40$, $FP = 10$, and $FN = 10$. Compute the overall classification accuracy.




**a.** 81.0%

**b.** 81.8%

**c.** 52.4%

**d.** 13

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

In a multiple linear regression model, the Sum of Squared Errors is $\mathrm{SSE} = 20$ and the Total Sum of Squares is $\mathrm{SST} = 100$. What is the coefficient of determination $R^2$?

**a.** $R^2 = 0.80$

**b.** $R^2 = 0.20$

**c.** $R^2 = 0.90$

**d.** $R^2 = 0.25$

**30** [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

In anomaly detection, anomaly prior $P(A) = 0.02$, detection sensitivity $P(+|A) = 0.95$, and false alarm rate $P(+|egA) = 0.05$. By Bayes' theorem, what is the posterior probability $P(A|+)$ that an alarm is a true anomaly?

**a.** 0.140

**b.** 0.950

**c.** 0.279

**d.** 0.391

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**31** **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

A 3D dataset has covariance matrix eigenvalues $\lambda_1 = 62$, $\lambda_2 = 24$, and $\lambda_3 = 14$. What percentage of total variance is explained by the first principal component $PC_1$?

**a.** 62.0%

**b.** 86.0%

**c.** 24.0%

**d.** 72.1%

**32** **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

Principal Component Analysis (PCA) projects a dataset from $\mathbb{R}^{50}$ onto its top $6$ principal components. What is the dimension of each transformed data point?

**a.** 25

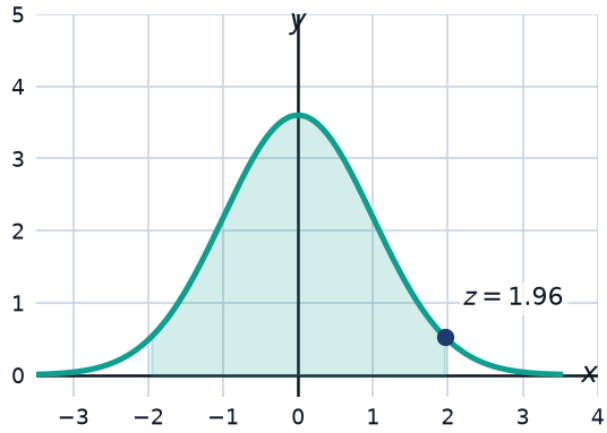
**b.** 50

**c.** 44

**d.** 6

**33** **[Intermediate]** • **Target: Gradient Descent & Optimization Practice Exercises**

A feature in an AI vision model follows a normal distribution $X \sim \mathcal{N}(25, 36)$. What is the mean $\mu$ and variance $\sigma^2$ of this feature?




**a.** $\mu = 25,\ \sigma^2 = 36$

**b.** $\mu = 36,\ \sigma^2 = 25$

**c.** $\mu = 25,\ \sigma^2 = 6$

**d.** $\mu = 6,\ \sigma^2 = 36$

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**34** [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

A logistic regression model produces a class probability $p = P(y = 1|\mathbf{x}) = 0.30$. Compute the corresponding log-odds (logit) $\ln\left(\frac{p}{1-p}\right)$.

**a.** 0.429

**b.** 0.353

**c.** $-2.347$

**d.** $-0.847$

**35** [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

A fitted simple linear regression model is $\hat{y} = 7x + 18$. What is the predicted target value $\hat{y}$ when the input feature is $x = 4$?

**a.** 46

**b.** 49

**c.** 42

**d.** 28

**36** [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

In a dataset, the prior probability of class $C_1$ is $P(C_1) = 0.10$. What is the probability of the complement class $P(egC_1)$?

**a.** 0.90

**b.** 0.95

**c.** 0.05

**d.** 1.10

**37** [Intermediate] • Target: Gradient Descent & Optimization Practice Exercises

Gradient descent step: $\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha ablaf(\mathbf{x}_n)$. For $f = x^2 + y^2$ at $(1, 1)$, $\alpha = 0.5$:

**a.** $(0.5, 0.5)$

**b.** $(0, 0)$

**c.** $(-1, -1)$

**d.** $(1.5, 1.5)$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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**38** [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Gradient descent update $\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha ablaf$ converges to:

**a.** a local min for smooth $f$

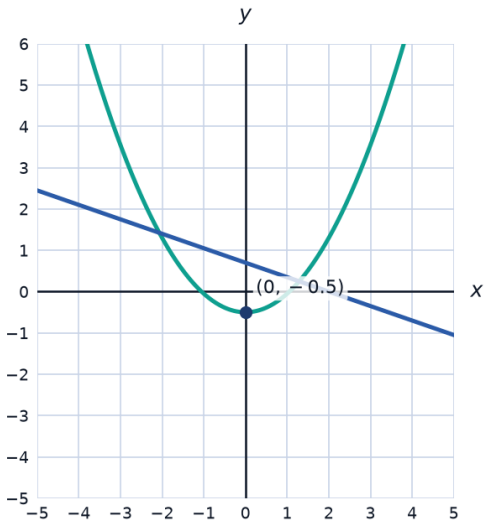
**b.** global min always

**c.** a saddle point

**d.** the origin

**39** [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Consider the function $f(x, y) = x^2 + y^2 - 4x - 6y + 13$. Find the coordinates of the critical point for this function.




**a.** (2, 3)

**b.** (-2, -3)

**c.** (4, 6)

**d.** (0, 0)

**40** [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Consider the function $z = f(x, y) = x^2 + y^2$. Let $x = 2t$ and $y = 3t$. Use the Multivariable Chain Rule to find $\frac{dz}{dt}$ when $t = 1$.

**a.** 10

**b.** 13

**c.** 26

**d.** 52

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**41** [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Consider the function $f(x, y) = x^3 + y^3 - 3axy$, where $a$ is a positive constant. It is known that $f(x, y)$ has a critical point at $(x_0, y_0)$ where $x_0 = 2$ and $y_0 = 2$. Find the value of $a$.

**a.** 1

**b.** 2

**c.** 3

**d.** 4

**42** [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

Consider the function $z = f(x, y) = x^2 + 2y^2$. What is the value of the partial derivative $\frac{\partial f}{\partial x}$ at the point $(1, 2)$?

**a.** 2

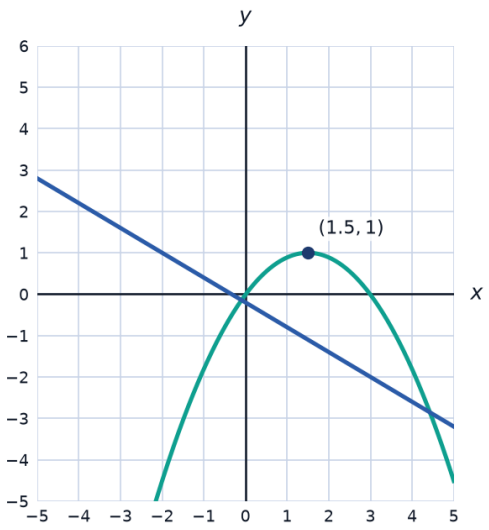
**b.** 4

**c.** 8

**d.** 10

**43** [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 5.01 Distributions and Maximum Likelihood

Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.




**a.** 2

**b.** 4

**c.** 6

**d.** 0

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**44** [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Consider the function $z = f(x, y) = x^2 - 3xy + 2y^2$. What is the partial derivative of $f$ with respect to $y$, denoted as $\frac{\partial f}{\partial y}$?

**a.** $2x - 3y$

**b.** $2x - 3xy + 4y$

**c.** $-3x + 4y$

**d.** $2x - 3y + 4y$

**45** [Challenging] • Target: Gradient Descent & Optimization Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

Let the scalar field be given by $f(x, y) = 3x^2 - 5y$. Compute the value of $f_x(2, -3)$.

**a.** 6

**b.** 12

**c.** -6

**d.** 15

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